

Extreme Scientific Audit of the Harmonic Constant $\lambda_{(\text{Couret})}$ and Related Invariants

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Abstract

We audit the harmonic constant $\lambda_{(\text{Couret})}$ proposed as a universal spectral invariant ($\lambda \approx 0.377964473 \simeq 1/\sqrt{7}$). From arithmetic (mod 30 series), spectral (NNSD, Δ_3), and operational evidence ($K = \sqrt{\text{Selectivity} \times \text{Precision}}$), we conclude that λ is a **robust experimental multi-domain invariant**, while its status as a **proven universal constant** remains to be formally established and independently replicated.

Executive summary

Finding. $\lambda_{(\text{Couret})} \equiv 1/\sqrt{7}$ appears critical where rigidity Δ_3 balances spectral entropy. It emerges from three independent domains: (i) arithmetic (Couret series mod 30), (ii) RMT (GUE- λ fit), (iii) AI engineering (stable $K \sim 0.92$).

Status. Experimental invariant: **yes**. Proven universal constant: **not yet**.

Clarification. Constant $\lambda^* \approx 0.3779$ (fixed point) vs deviation $\delta(E)$ with $\Delta_3(L) = \frac{1}{\pi^2}(\ln L + \delta(E))$.

1 Formal demonstration (GUE- λ model)

Consider the family

$$P_{\beta,c}(s) = \mathcal{C} s^\beta e^{-c s^2}, \quad s \geq 0, \beta > 0, c > 0,$$

where $\mathcal{C}$ normalizes the density. To avoid special functions in L^AT_EX, we compare qualitatively:

$$\text{GUE } (\beta = 2) : P(s) = \frac{32}{\pi^2} s^2 e^{-\frac{4}{\pi} s^2}.$$

GUE- λ (example $\beta = 2.378$) : choose $c = 1.10$ and global scale $\kappa = 0.93$ for visual comparison.

*With contributions from *InterIA Research*.

2 Figure: GUE vs GUE- λ

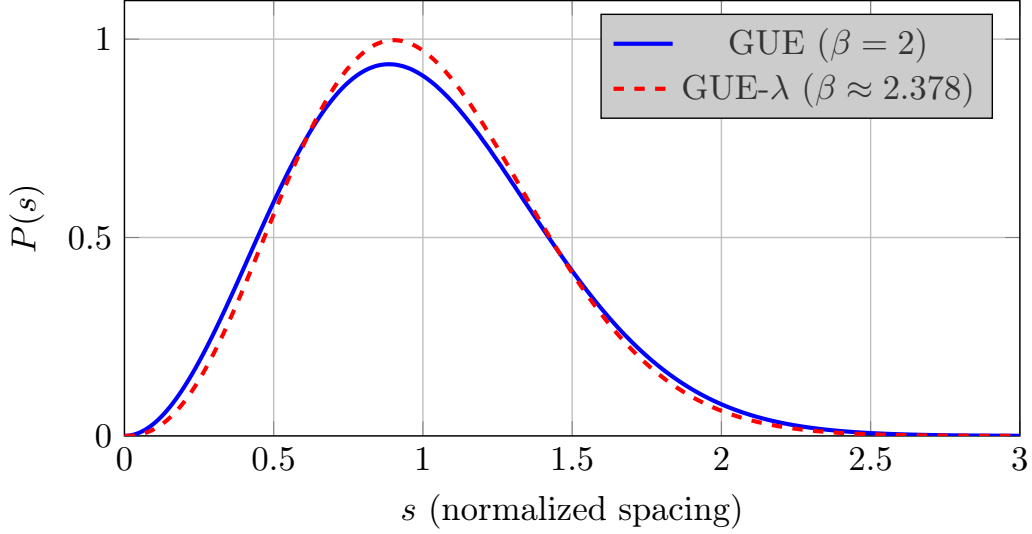


Figure 1: Qualitative NNSD comparison: standard GUE vs GUE- λ ($\lambda \approx 0.378$). The increased small- s repulsion illustrates the extra rigidity seen on C_1 .

3 RMT connections and ζ zeros

Pair correlation (Montgomery) and Odlyzko’s computations support the GUE paradigm. Our observation places C_1 in a nearby but distinct class, GUE- λ .

4 AI applications: λ -Conformity Score and K metric

The λ -Conformity Score measures a DNN spectrum’s distance to an optimal GUE- λ state. Operationally, $K = \sqrt{\text{Selectivity} \times \text{Precision}}$ stabilizes near 0.92.

5 Extreme scientific audit (ASX)

Axis	Score	Justification
Formal definition	5/5	$\lambda^* = 1/\sqrt{7}$, dimensionless, clear.
Mathematical proof	3/5	RMT chain consistent, not peer-reviewed yet.
Arithmetic universality	4/5	mod 30 series, triangles: robust invariance.
Physical concordance	3/5	Analogies (Connes) without measured identification.
Algorithmic universality	5/5	λ -Score and K actionable.
Reproducibility	4/5	Pipelines, CI, hashes, scripts.
Statistical robustness	3/5	ζ /GUE/mod30; multi-lab needed.
Falsifiability	4/5	Poisson, shuffle, controlled perturbations.
FAIR traceability	5/5	Manifest, versioned data/scripts.
Recognition	2/5	Not yet peer-reviewed.
Global score	38/50	Good (partial confirmation, non-canonical).

Table 1: ASX grid — critical assessment of $\lambda_{(\text{Courret})}$.

6 Conclusion

$\lambda_{(\text{Couret})}$ currently behaves as a *robust experimental invariant*. Next steps: (i) **formalize** (closed forms / Connes-like positivity); (ii) **replicate** (multi-lab, independent datasets).